Robot Manipulator – mechanical arm used to perform tasks such as picking objects, moving materials and collecting samples. It has links that allow different parts of arm to move independently.

DOF – No. of independent movements a robotic mechanism can perform. Each independently controlled joint adds to DOF.

Revolute joint – All rotational movement around a fixed axis (movement of elbow)
Prismatic joint – Allow linear movement along a straight path

X-axis → forward & backward
Y-axis → Left & Right
Z-axis → Up & Down
Roll → Rotate around X-axis
Pitch → Rotate around Y-axis
Yaw → Rotate around Z-axis

DOF w/ & w/o Gripper

Total no. of DOF depend on no. of independently moving joint in robotic arm.

The gripper must also be considered but it may have its own movement mechanism. (eg- open close gripper add one independent DOF)

	DOF
Robotic arm w/o gripper	5
" " w/ single motor gripper	6
" " w/ 2 indepnt gripper movement	7

These values are only example. The actual DOF of a manipulator must be calculated by examining its joints. A gripper w/ 2 fingers connected to same motor generally counts as one DOF bcz both fingers move together.

It is also help to understand that AI does not make physical work. It helps the robot decide how to use its existing parts in a task.

Forward kinematics - Used to calculate the position and orientation of the gripper using the known joint angles and link length of the robotic arm.

↳ We know how much each joint has to rotated, forward kinematics tells us where the gripper will reach.

$$x = L_1 \cos\theta_1 + L_2 \cos(\theta_2 + \theta_1)$$
$$y = L_1 \sin\theta + L_2 \sin(\theta_2 + \theta_1)$$

Inverse Kinematics - works in opposite way to forward kinematics. Instead of calculating the final position from joint angles, it calculates the joint angles required to reach a particular position.

Difference

forward	Inverse
• Uses joint angle as input	• Uses target position as input
• Calculates the gripper's position	• Calculates the required joint angles
• Used to track arm's position	• Used to move arm towards target

Role of AI in manipulator

AI can make a robot manipulator more useful by helping it recognize objects & decide what action needs to be performed.

AI can also help w/ object classification, target selection & motion planning. However, the actual joint movement are calculated using kinematics eqn & controlled through robot's motor system.

for Ex-

1) A camera capturing surrounding area.
2) An AI model identifies the rock that needs to be collected.
3) Stereo vision or another depth sensor estimates its position in three-dimensional space.
4) Inverse kinematics calculates the joint angle required to reach rock
5) The gripper closes around rock, sensors help confirm whether it has been picked up successfully.

Conclusion

DoF determines how many independent movement a robotic manipulator can perform. Forward kinematics calculates the position of gripper from joint angles, while inverse kinematics determine joint angle needed to reach a particular position.

when these components are combined w/ AI & CV, a rover can identify objects, calculate their position & perform tasks such as collecting soil & rock samples w/ limited human intervention.